

NEWPORT, VT, USA

WMO: 720493

Lat: 44.883N

Lon: 72.233W

Elev: 915

StdP: 14.22

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 54758

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-6.0	-19.3	2.0	-9.2	-14.8	2.5	-3.6	23.3	27.5	20.4	23.8	3.6	220	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	84.2	69.2	81.4	66.4	79.0	66.0	72.0	79.7	70.3	77.5	68.5	75.6	8.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.5	112.2	76.0	67.8	105.8	74.5	65.9	99.0	72.7	36.3	80.0	34.8	77.5	33.2	75.5	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	16.6	14.5		-19.0	88.9	4.9	2.1	-22.5	90.5	-25.4	91.7	-28.1	92.9	-31.6	94.4
			DB	-19.4	74.6	4.5	1.9	-22.6	75.9	-25.3	77.1	-27.8	78.1	-31.1	79.5
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	16.5	19.7	27.1	41.1	54.1	61.3	67.2	65.5	58.6	46.7	34.6	23.9		
	DBStd	19.93	12.49	11.54	11.39	9.06	8.14	6.68	5.44	5.14	7.65	8.49	9.83	11.59		
	HDD50	4404	1038	849	713	296	48	3	0	0	12	168	467	810		
	HDD65	8230	1503	1269	1174	717	349	149	39	57	220	567	911	1275		
	CDD50	1907	0	0	4	30	176	342	534	480	269	66	6	0		
	CDD65	260	0	0	0	1	13	39	108	72	27	0	0	0		
	CDH74	2058	0	0	4	18	126	344	858	492	211	5	0	0		
	CDH80	394	0	0	0	3	17	74	189	73	38	0	0	0		
	WSAvg	5.8	6.1	6.2	7.0	7.3	6.1	5.2	4.6	4.6	4.8	5.8	6.1	5.5		
	PrecAvg	40.0	2.4	2.3	2.7	3.0	3.5	4.1	4.0	4.4	3.6	3.2	3.6	3.2		
(15) Precipitation	PrecMax	52.5	6.5	4.4	4.3	5.7	6.7	10.7	7.2	7.5	7.8	7.0	6.6	7.1		
	PrecMin	30.6	0.5	0.4	0.7	0.9	1.2	1.7	1.5	1.8	1.1	0.7	1.9	1.6		
	PrecStd	5.9	1.4	1.0	0.9	1.1	1.6	2.1	1.5	1.5	1.7	1.3	1.1	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.2	50.4	73.0	76.6	82.2	87.3	88.3	85.9	84.4	72.9	63.7	50.3		
		MCWB	44.3	45.4	58.1	58.7	66.1	69.7	70.2	69.7	71.8	61.1	53.5	46.3		
	2%	DB	41.6	43.4	53.6	67.9	77.4	81.8	84.9	81.7	79.5	67.8	57.9	45.9		
		MCWB	38.2	38.4	44.8	54.8	62.0	65.5	69.9	66.7	67.0	57.6	50.8	43.0		
	5%	DB	37.3	39.4	46.4	62.5	73.1	78.5	81.6	79.2	76.1	63.7	53.9	42.5		
		MCWB	34.1	34.9	39.9	51.4	59.8	64.1	67.2	66.7	65.6	56.0	47.7	39.6		
	10%	DB	33.9	36.1	42.6	55.7	69.6	74.0	78.9	76.8	72.3	60.4	49.2	37.8		
		MCWB	31.2	32.8	36.6	46.2	57.6	62.8	66.8	66.0	63.7	54.3	44.5	35.1		
	0.4%	WB	45.1	47.6	57.5	61.1	68.7	72.2	75.4	73.0	72.4	65.1	56.1	46.9		
		MCDB	46.1	48.4	72.4	70.8	79.0	81.2	84.6	80.5	80.6	68.1	59.5	49.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.2	39.6	45.6	57.3	65.6	69.5	72.6	70.7	69.8	60.4	52.4	43.8		
		MCDB	41.5	42.6	51.4	66.2	74.0	77.4	79.9	77.8	76.9	64.3	56.9	45.9		
	5%	WB	34.7	35.7	40.8	52.1	62.5	67.1	70.8	68.8	67.6	57.7	48.2	39.9		
		MCDB	37.4	39.0	46.0	59.0	69.9	75.3	78.0	75.2	74.4	63.6	52.5	42.4		
	10%	WB	31.6	32.8	37.2	47.6	59.6	64.7	68.8	67.4	65.1	54.7	44.2	35.8		
		MCDB	33.4	35.2	41.8	56.1	67.0	71.3	75.9	74.1	70.8	59.3	48.7	38.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	16.6	16.7	17.7	20.4	18.9	19.3	19.0	19.6	16.0	13.8	13.1		
		MCDBR	15.4	17.9	20.4	27.6	25.1	24.4	22.8	22.0	23.4	20.9	21.1	15.7		
		MCWBR	14.7	14.6	13.7	16.4	13.7	11.6	10.7	10.9	12.9	13.5	15.1	14.5		
		MCDBR	15.1	17.5	18.5	24.0	20.8	19.4	19.3	18.7	21.5	18.5	18.7	14.8		
	5% WB	MCWBR	14.7	15.3	13.6	15.9	13.2	10.6	10.7	10.7	13.1	13.6	15.6	14.5		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.281	0.317	0.349	0.367	0.416	0.425	0.398	0.361	0.324	0.304	0.272		
	taud		2.389	2.299	2.334	2.385	2.396	2.249	2.238	2.304	2.406	2.514	2.493	2.408		
	Ebn at Noon		277	288	291	289	286	271	267	270	272	270	257	258		
	Edh at Noon		23	31	34	36	36	43	43	38	32	25	21	21		
(44) All-Sky Solar Radiation	RadAvg		448	730	1098	1355	1576	1728	1748	1568	1259	734	452	323		
	RadStd		56	81	78	163	170	145	136	81	98	75	60	28		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+117	N/A

Nomenclature: See separate page